

Multi-source comparison of recent terrestrial evapotranspiration trends: Introducing a topology framework

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Why this matters

Evapotranspiration (ET) couples terrestrial water, energy, and carbon exchanges. Global ET products are widely used to infer recent hydroclimatic change, but trend estimates remain strongly product dependent in magnitude, spatial pattern, statistical significance, and even direction.

Research gap: conventional ensemble spread identifies disagreement, but does not show *which products amplify, dampen, or oppose* the ensemble trend signal.

Datasets and harmonisation

- **14 global ET products**, annual ET totals for **2000–2019**.
- Harmonised to a common **0.25° × 0.25°** grid.
- Ensemble diagnostics calculated only where product coverage is sufficient for robust comparison.

Product type	Examples
Reanalysis	ERA5-Land, JRA-55, MERRA-2
Remote sensing	BESS v2, ETMonitor, GLEAM v4.1a, MOD16A2
Land surface models	FLDAS, GLDAS-CLSM, GLDAS-NOAH, GLDAS-VIC, TerraClimate
Composite/synthesis	CAMELE, SynthesizedET

Trend estimation

- Grid-scale and regional ET trends estimated with a **block-bootstrapped Theil–Sen estimator**.
- Block length: **3 years**; bootstrap repeats: **599**.

Global trend agreement and uncertainty

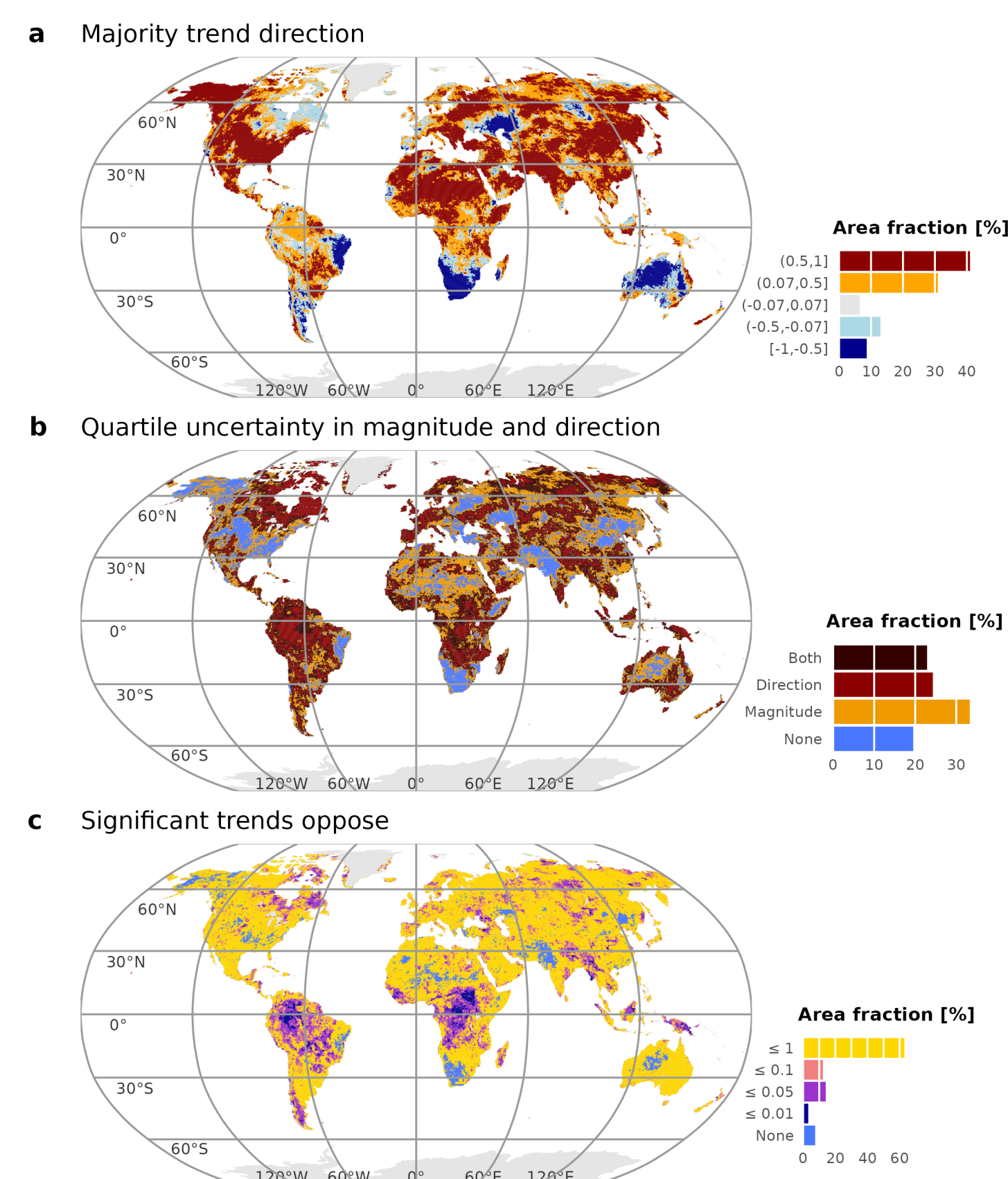


Figure 1: Variation in ET trends across products. (a) Majority trend direction irrespective of significance (red: positive, blue: negative, gray: equal positive and negative trends). (b) Quartile uncertainty in trend magnitude and direction. (c) P-value indicating onset of directional opposition among significant trends.

- Positive ET trends dominate the ensemble, but agreement is not uniform.
- Large-scale trend signals can be driven by a limited number of products rather than broad inter-product concurrence.
- Significant opposition occurs even where products indicate strong trends.
- Trend direction and magnitude remain highly product dependent.

Aim

Introduce a reproducible **topology framework** that classifies ET products by their trend signatures and ensemble roles, enabling structured comparison of product-specific uncertainty and agreement patterns.

Topology framework

Each product is ranked by the area fraction associated with six interpretable roles. This converts multi-product trend maps into a compact taxonomy of ensemble behavior.

Role	
Positive signal booster	Product has a large area fraction of significant positive trends.
Negative signal booster	Product has a large area fraction of significant negative trends.
Signal dampener	Product has a large area fraction of non-significant trends.
Trend opposer	Product often has the opposite trend direction to the ensemble majority.
Significance opposer	Product often differs from the majority in trend significance.
Opposition contributor	Removing the product removes directional opposition in significant trends.

Product roles in global topology

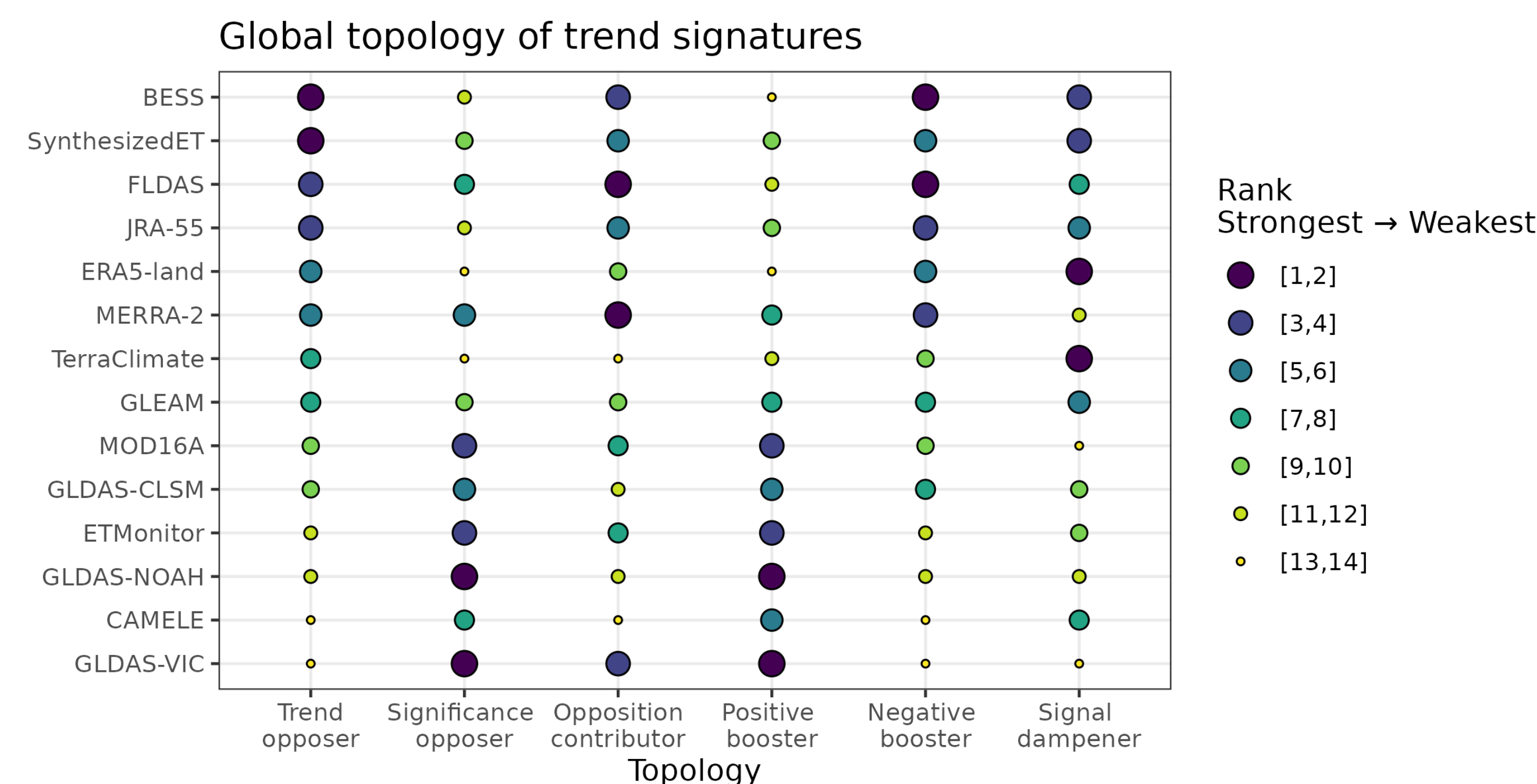


Figure 3: Global topology of trend signatures. Darker color and larger circles indicate a stronger signature of a given role. Datasets are ordered according to their rank as

“Trend opposers”, highlighting products that most frequently deviate from the majority trend direction across the ensemble.

- **Negative signal boosters often act as top outliers.** These products generate significant negative trends where most other significant trends are positive.
- **Positive signal boosters often act as significance opposers.** They produce significant positive trends where most products show non-significant trends.
- **Opposition is structured, not random.** Product roles persist across p-value thresholds, although role strength varies.
- **Regional context matters.** A product with moderate global behavior may still strongly oppose the regional majority trend.

Conceptual topology categories

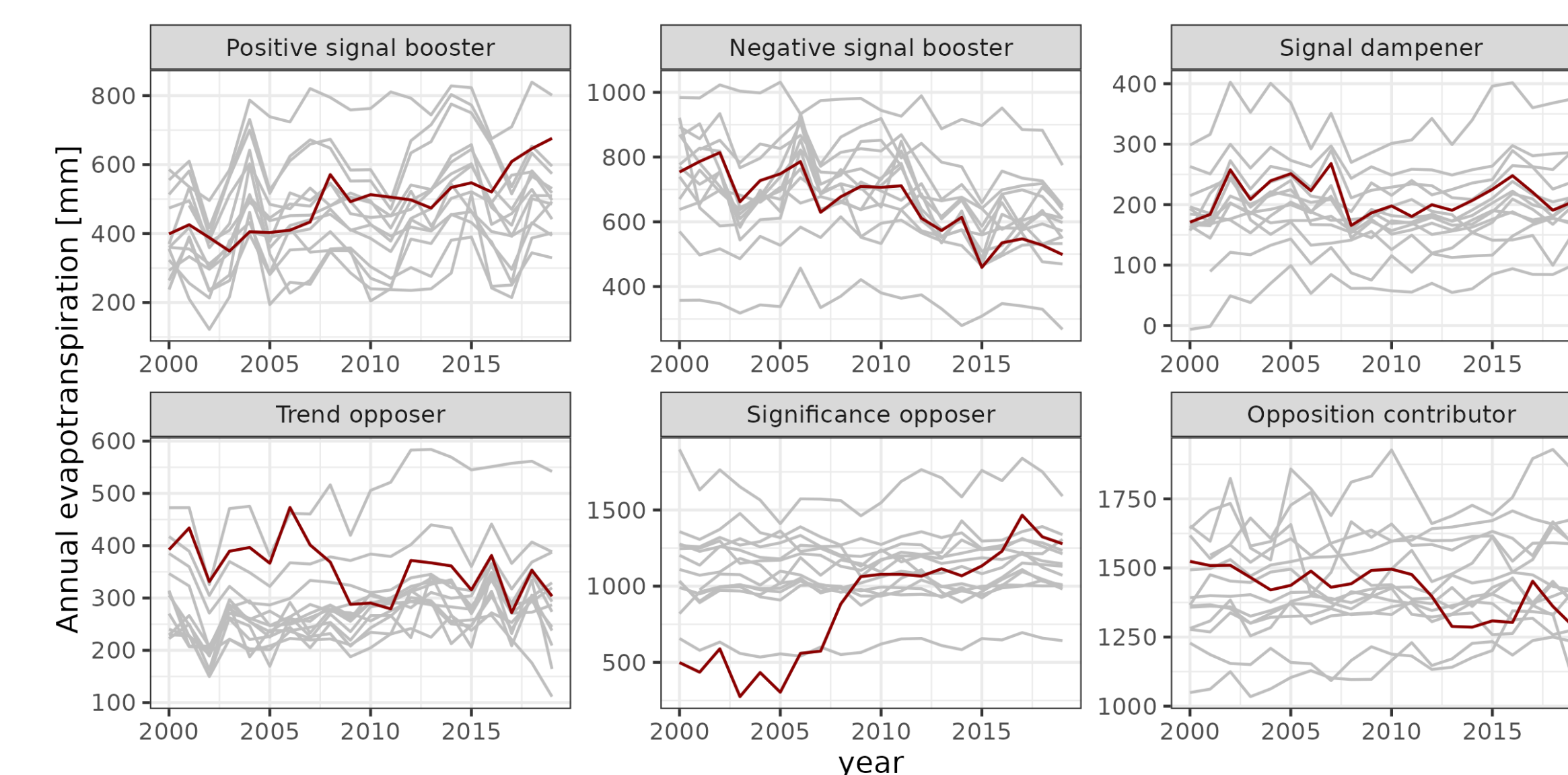


Figure 2: Time-series behavior for signal boosters, dampeners, trend and significance opposers, and opposition contributors.

Use-case: Regional opposition in GLEAM

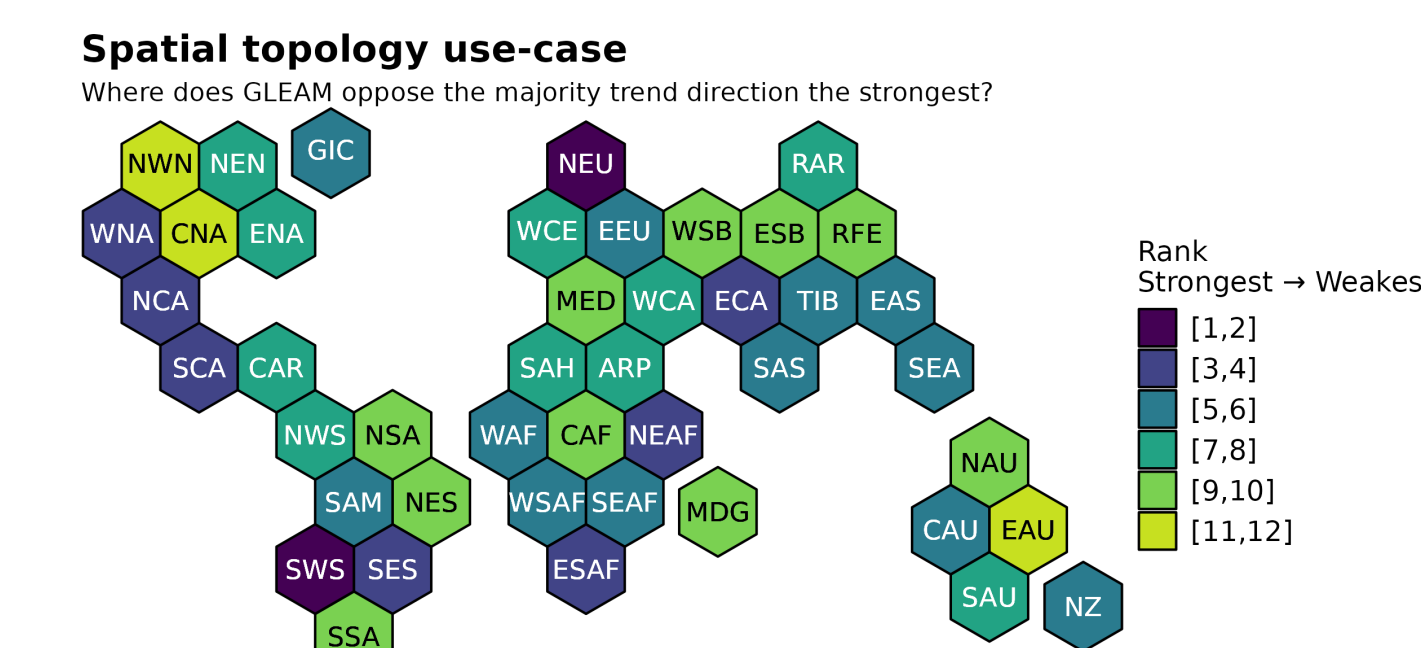


Figure 4: Where does GLEAM 4.1a oppose majority trend direction the most?

- Strongest opposition occurs in NEU and SWS.
- GLEAM appears globally moderate but regionally divergent.
- Single-product ET interpretation may become dataset dependent.

Take-home message

ET trend uncertainty should be interpreted not only as ensemble spread, but as a structured property of multi-product ensembles. The topology framework identifies which products amplify, dampen, or oppose trend signals, supporting more transparent product selection and uncertainty assessment.

Funding, Contact and code

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